

Roll No.

TCS-503

**B. TECH. (CSE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

DATA BASE MANAGEMENT SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Discuss the main characteristics of database approach. How does it differ from traditional database ? (CO1)

OR

- (b) Discuss the concept of data independence and explain its importance in a database environment. Also discuss the referential integrity constraints in relational databases. (CO1)

2. (a) What are the responsibilities of a DBA? If we assume that the DBA is never interested in running his or her own queries, does the DBA still need to understand query optimization ? Why ? (CO1)

OR

- (b) Which language is used in application programs to retrieve, store, modify, delete, insert and update in database ? Discuss about that language with the operations of application programs to retrieve, store, modify, delete, insert and update in database. (CO1)

P. T. O.

3. (a) Draw an E-R Diagram for below scenario. Take assumptions if required.

A Blood Bank stores blood of various blood groups. Many donors donate blood, each of different blood group/type. A donor may donate blood more than once and he is identified by a donor id (DID), name, Gender, age, address and phone number. The blood donated by the donor is characterized by blood type, code and cost. Before each donor donates his blood, he is required to register himself as a donor with the receptionist who works at the Blood Bank. The receptionist is identified by employee id, name, address and phone number. The Blood Bank receives orders for blood from many hospitals for emergency purposes and other surgical requirement and each blood bank issues the same of required blood type. Each blood bank has its own blood bank number (BNO), issues, orders and blood types stored. The Blood Bank is managed by the blood bank manager who is identified by employee id, name, email id and phone number. He is responsible for the proper management of the blood bank. The hospitals are identified by name, address and phone number. (CO2)

OR

- (b) Differentiate between of the following : (CO2)

- (i) Weak entity set and strong entity set.
- (ii) Total participation and partial participation
- (iii) Specialization and generalization
- (iv) Multivalued attribute and composite attribute

4. (a) Why are data models required in database management systems ? Explain types of data models with their advantages and disadvantages. (CO2)

OR

- (b) Why is data abstraction important in DBMS ? Explain each level of data abstraction with suitable example. (CO2)
5. (a) Consider the below schema and solve the relational algebra for the following statements : (CO3)

CUSTOMER (account_no, Computer_name, Customer_city)

LOAN (loan_no, Branch_name, amount)

DEPOSITOR (Customer_name, account_no)

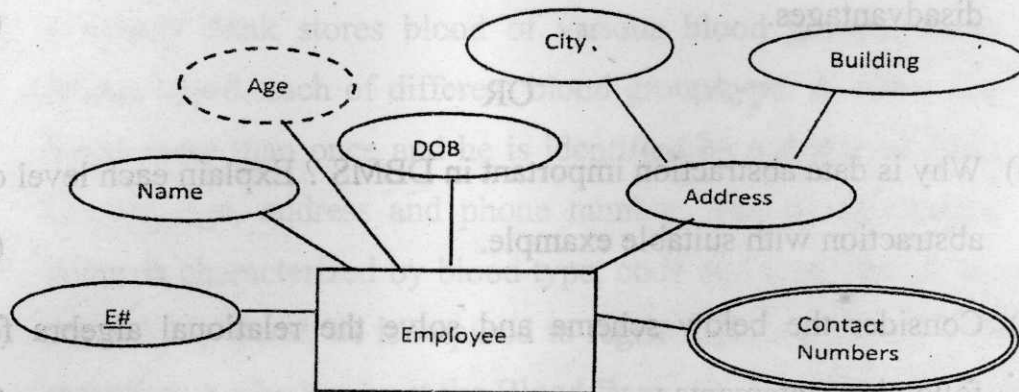
BORROWER (customer_name, loan_no)

- (i) Find all tuples of the relation customer who lives in the city 'Dehradun'.
- (ii) Find those customer names who have either a loan or account in the bank.
- (iii) Find the names of all the customers who have loan at the 'AXIS-Vikas Nagar' branch.
- (iv) Find all the loan numbers of 'AXIS—Vikas Nagar' branch where the amount is more than 12000.
- (v) Find those customer names who have loan of amount 15000 in the bank.

P. T. O.

OR

(b) Consider the below E-R diagram and solve the below SQL queries :



- (i) Create the employee table.
- (ii) List the name of employee who belongs to Dehradun and salary is greater than 10000.
- (iii) List the employees who are not live in Delhi.
- (iv) List the emp. names those are having seven character and fifth char. must be 'U'.
- (v) Modify the salaries of all employees by giving 10% raise in their salaries.